SODIUM STIBOGLUCONATE

Cardiotoxic — ECG monitoring

Endemic in Iraq

CHEMICAL NAME Pentavalent antimony (Sb^{5+}) complex of sodium gluconate (100 mg Sb^{5+}/mL)

CLASS Pentavalent antimonial antiprotozoal (antileishmanial)

TRADE NAMES Pentostam · Glucantime (meglumine antimoniate — similar)

THERAPEUTIC USES / INDICATIONS

- Cutaneous leishmaniasis ("Baghdad boil", oriental sore); intralesional or systemic.
- Visceral leishmaniasis (kala-azar).
- Mucocutaneous leishmaniasis.

PHARMACODYNAMICS

- Inhibits *Leishmania* **glycolysis and fatty-acid oxidation**, and topoisomerase → reduced ATP/GTP → parasite death.
- Reduced to trivalent antimony (active form) in amastigotes.

ADMINISTRATION

- **Intralesional:** infiltrate base and edges of lesion until blanched, weekly or every few days.
- **IV:** slowly over ≥5 min (or diluted infusion) — via fine needle to reduce thrombosis; **IM** (painful).
- Doses expressed as mg of antimony (Sb^{5+}).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Myalgia, arthralgia, fatigue.
- Nausea, anorexia, abdominal pain.
- Injection pain; local reactions with intralesional use.

ADVERSE EFFECTS (SERIOUS)

- Cardiotoxicity: QT prolongation, T-wave changes, arrhythmias, sudden death.
- Pancreatitis (raised amylase/lipase).
- Hepatotoxicity; nephrotoxicity.
- Bone-marrow suppression.

DOSAGES

- **Systemic:** 20 mg Sb^{5+}/kg/day IV or IM — 20 days (cutaneous) or 28–30 days (visceral/mucosal).
- **Intralesional:** 0.5–2 mL per lesion every 3–7 days for several sessions.
- Same dose in children.

CONTRAINDICATIONS & CAUTIONS

- Significant cardiac disease, long QT.
- Significant renal or hepatic impairment.
- Pregnancy (relative), breastfeeding.
- Hypersensitivity.

KEY INTERACTIONS

- **QT-prolonging drugs** (antiarrhythmics, macrolides, antipsychotics): avoid.
- **Amphotericin B:** give sequentially, not concurrently (cardiotoxicity).

PHARMACOKINETICS

- **Onset:** IV immediate; IM peak 2 h.
- Not metabolised.
- Excreted rapidly in urine (most within 24 h); slow terminal phase. **t½:** 2 h (initial).

NURSING MANAGEMENT

- Baseline and twice-weekly ECG (QTc) during systemic therapy.
- Weekly CBC, LFTs, renal function, amylase/lipase.
- Monitor for abdominal pain (pancreatitis), palpitations.
- Measure and photograph lesions to track healing.
- Encourage fluid intake; manage myalgia.

PATIENT EDUCATION

- Protect lesions from injury and infection; keep clean and covered.
- Report palpitations, fainting or severe abdominal pain.
- Healing may continue for weeks after treatment ends.
- Use insect repellent and bed nets to avoid sandfly bites.

KEY POINTS

- Pentavalent antimony for leishmaniasis (Baghdad boil).
- Cardiotoxic — monitor ECG/QT.
- 20 mg Sb^{5+}/kg/day systemic.
- Pancreatitis and hepatotoxicity.